

Mobolaji Ogunbiyi

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EDUCATION

University of Maryland, Baltimore County (UMBC)

Baltimore, MD

B.S. in Computer Science, B.A. in Psychology

Expected May 2027

GPA: **3.59/4.00** | Meyerhoff Scholar | President's List | Semester Academic Honors

Selected Coursework: Data Structures, Operating Systems, Computer Architecture, Linear Algebra, Statistics, Discrete Structures, Programming Languages

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Swift, SQL

Frameworks/Tools: FastAPI, Flask, Django, React, Vite, Git, Postman, Jupyter, Xcode

Libraries/Platforms: NumPy, Pandas, scikit-learn, TensorFlow, OpenCV, SQLite, FTS5, sqlite-vec, pgvector, Google Cloud, Firestore, AlloyDB

AI/Agents: LangGraph, prompt engineering, agent orchestration, tool-calling agents, long-horizon task execution, memory systems, vector search, LLM evaluation, NLP

EXPERIENCE

M&T Bank - Technology Intern

June 2025-August 2025

- Built an internal record-mapping tool with a Flask REST API and React/Vite frontend to reduce manual retention-code lookup work for the DDRR team.
- Developed an NLP pipeline that mapped document descriptions to retention codes with **80%+ top-1 accuracy**, comparing TF-IDF and ConceptNet Numberbatch approaches with cosine similarity evaluation.
- Drove early adoption by demoing the tool to stakeholders across the organization and helping position it for internal onboarding.

JUNEBRAIN Inc. - Software Engineering Intern

January 2025-May 2025

- Built a Python/OpenCV pupil-localization pipeline for Junebrain's Neuro-i OCT device to detect gaze misalignment and flag low-fidelity frames in real time.
- Implemented ellipse-geometry and thresholding logic for 2D gaze estimation across multi-camera input streams.
- Evaluated and adapted ML-based gaze models, including GazeML and NVIDIA TAO GazeNet, to improve eye-only inference and TensorFlow 2 compatibility.

UMBC - Undergraduate Researcher

October 2024-Present

- Maintained Python web-scraping pipelines to collect smart-IoT-device metadata and privacy policies, and built comparison tooling to detect newly added or updated policies across runs.
- Automated privacy-policy extraction and analysis with LLM- and NLP-based workflows, contributing experimental results and writing support for an ongoing research paper.
- Built reproducible analysis scripts to support validation, readability scoring, and large-scale policy comparison across repeated data-collection sessions.

PROJECTS

MoLabel - Founder

February 2025-Present

- Built a local-first AI data operations workbench for agent-assisted labeling, review, and dataset release workflows using a Python/FastAPI backend, React frontend, and Tauri desktop shell.
- Designed evaluation and benchmarking pipelines to measure agent performance on real-world data tasks, including labeling quality, failure modes, auditability, and human-in-the-loop correction paths.
- Added observability and review tooling to compare agent outputs, route edge cases to human reviewers, and track quality shifts across workflow runs.

Kora - Local-First Autonomous AI Agent System

November 2025-Present

- Architected an ADHD-aware local-first AI runtime in Python with a FastAPI daemon, WebSocket/REST interfaces, and a LangGraph supervisor coordinating planner, executor, and reviewer agents.
- Implemented system-level execution controls for low-friction autonomous workflows, including time-bounded micro-steps, context compaction, interruption handling, checkpointing, and budget enforcement.
- Built persistent hybrid memory with filesystem-backed storage, SQLite, FTS5, and sqlite-vec to support recall and continuity across sessions; added typed contracts and quality gates to improve reliability.

LEADERSHIP & ACTIVITIES

TechBuffalo PowerUpTech 2025 Design Challenge - 1st Place

July 2025

Led UI/UX and system design for *Seneca Signal*, a community information platform combining a website, admin dashboard, and SMS distribution workflow; later presented the project at TechConnect 2025.

Computer Science Tutor (Python, C++) | UMBC NSBE | UMBC African Student Association